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15. An organ pipe A closed at one end is allowed to vibrate in its first harmonic and another pipe B open at both ends is allowed to vibrate in its third harmonic. Both A and B are in resonance with a given tuning fork. The ratio of the length of A and B is

(a) $\frac{8}{3}$ (b) $\frac{3}{8}$

(c) $\frac{1}{6}$ (d) $\frac{1}{3}$

Answers:

1) b	2) d	3) b	4) a
5) c	6) a	7) d	8) c
9) a	10) a	11) d	12) a
13) b	14) d	15) c	

II. Short Answer Questions

1. What is meant by waves?.
2. Write down the types of waves.
3. What are transverse waves?. Give one example.
4. What are longitudinal waves?. Give one example.
5. Define wavelength.
6. Write down the relation between frequency, wavelength and velocity of a wave.
7. What is meant by interference of waves?.
8. Explain the beat phenomenon.
9. Define intensity of sound and loudness of sound.
10. Explain Doppler Effect.
11. Explain red shift and blue shift in Doppler Effect.
12. What is meant by end correction in resonance air column apparatus?
13. Sketch the function $y = x + a$. Explain your sketch.
14. Write down the factors affecting velocity of sound in gases.
15. What is meant by an echo?. Explain.

III. Long Answer Questions

1. Discuss how ripples are formed in still water.
2. Briefly explain the difference between travelling waves and standing waves.
3. Show that the velocity of a travelling wave produced in a string is $v = \sqrt{\frac{T}{\mu}}$
4. Describe Newton's formula for velocity of sound waves in air and also discuss the Laplace's correction.
5. Write short notes on reflection of sound waves from plane and curved surfaces.
6. Briefly explain the concept of superposition principle.
7. Explain how the interference of waves is formed.
8. Describe the formation of beats.
9. What are stationary waves?. Explain the formation of stationary waves and also write down the characteristics of stationary waves.
10. Discuss the law of transverse vibrations in stretched strings.
11. Explain the concepts of fundamental frequency, harmonics and overtones in detail.
12. What is a sonometer?. Give its construction and working. Explain how to determine the frequency of tuning fork using sonometer.